CREDIT CARD SPENDING BEHAVIOUR

A Report Submitted in

Partial Fulfilment for

award of Bachelor of Technology

for

DATA ANALYTICS (ACSDS0651)

In

COMPUTER SCIENCE & ENGINEERING

By

STUTI (2301331520189)

Under the Supervision of

Ms. Ayushi Gupta

Assistant Professor

Department of Computer Science & Engineering

School of Computer Science and Information Technology

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY

GREATER NOIDA

(An Autonomous Institute)

Affiliated to

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

CERTIFICATE

This is to certify that the project titled"**Credit Card Spending Behavior Using Tableau"** submitted by **Stuti** is a bonafide work carried out during the Data Analytics Workshop.

Faculty Signature

Ms. Ayushi Gupta

Date: 17/03/2026

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our faculty mentor **Ms. Ayushi Gupta** for her valuable guidance, support, and continuous encouragement throughout the development of this project. Her suggestions and supervision greatly helped us in completing this work successfully.

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ABSTRACT

This project focuses on analyzing credit card transaction data in India to understand customer spending behavior. With the increasing use of digital payments, analyzing transaction data helps identify trends, patterns, and user preferences.

The dataset contains details such as transaction amount, category of spending, location, and transaction type. Tableau was used for data visualization and analysis.

Exploratory Data Analysis (EDA) techniques were applied to identify spending patterns across different categories, cities, and time periods. The analysis highlights major spending sectors, peak transaction periods, and customer behavior trends.

The results of this project help businesses, banks, and financial institutions make better decisions related to marketing strategies, fraud detection, and customer targeting.

INTRODUCTION

With the rapid growth of digital payments, credit cards have become a major mode of financial transactions. Every transaction generates valuable data that can be analyzed to understand consumer behavior.

Analyzing credit card data helps in identifying spending patterns, popular categories, and regional trends. This information is useful for banks, businesses, and policymakers.

This project uses Tableau to analyze credit card transaction data in India and extract meaningful insights using data visualization techniques

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OBJECTIVE

The main objectives of this project are:

- To clean and preprocess the credit card transaction dataset
- To analyze customer spending behavior
- To identify popular spending categories
- To visualize transaction trends using Tableau
- To extract insights for better decision-making

DATASET DESCRIPTION

Dataset Name: Credit Card Spending Behaviour

Source: Card Transactions – India Dataset (CSV file)

Variable	Description	Data Type
transaction_id	Unique ID of transaction	Categorical
amount	Transaction amount	Numeric
city	City where transaction occurred	Categorical
date	Date of transaction	Date
category	Type of expense (food, travel, etc.)	Categorical
card_type	Type of credit card	Categorical
gender	Customer gender	Categorical

METHOCOLOGY

Tools / Technologies Used:

Tableau Desktop

Used for data visualization, dashboard creation, and analysis.

Data Visualization Techniques

Bar charts, pie charts, line charts, and heatmaps for spending patterns.

Dashboard Design

Interactive dashboards combining spending by category, gender, segments.

Data Cleaning & Preprocessing

Before performing the analysis, the dataset was reviewed and prepared to ensure data quality and accuracy.

The following steps were performed:

- Removed missing or null values
- Removed duplicate records
- Converted date column into proper format
- Categorized spending types properly
- Filtered irrelevant or inconsistent data

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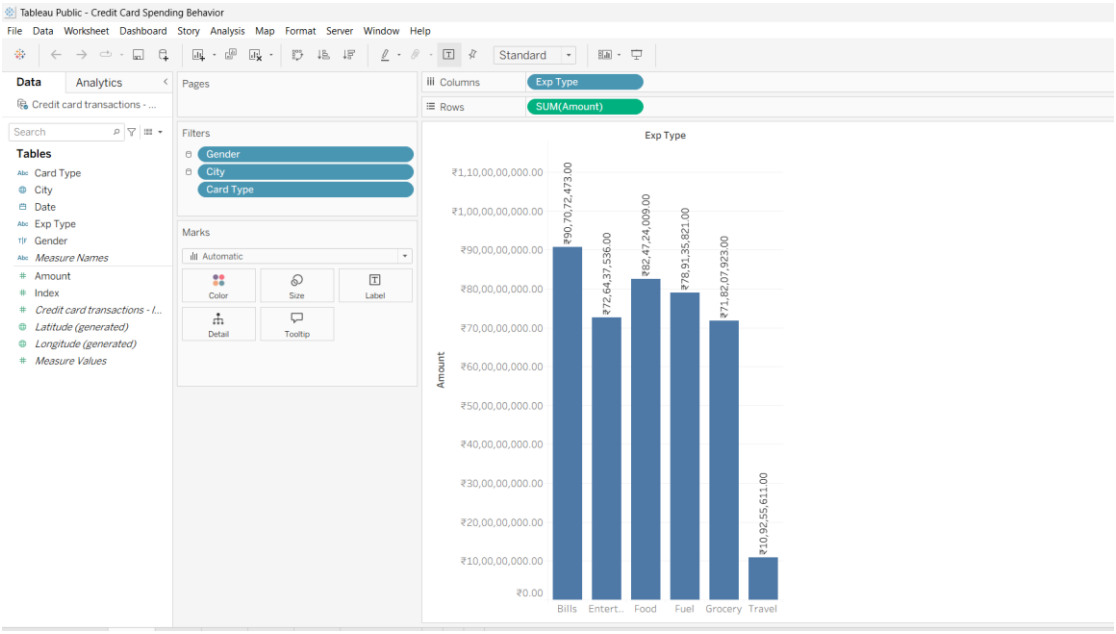
Exploratory Data Analysis (EDA)

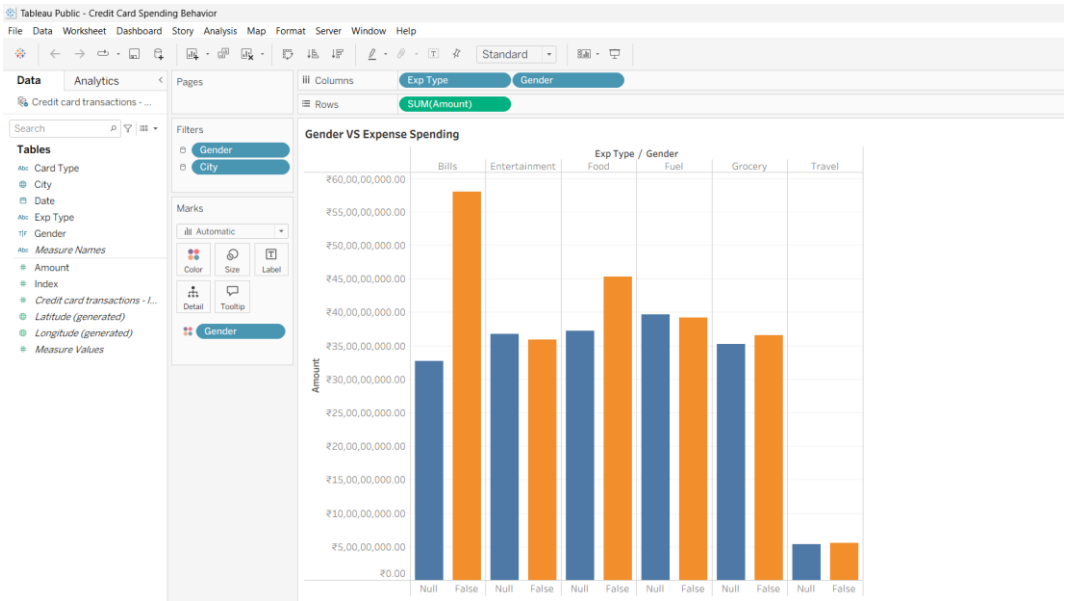
1. Spending by Catefory

Highest in shopping, food via stacked bars.



2. Spending by City/Segment

Top cities and segments shown.



3. Gender vs Spending

Comparison bars for male/female

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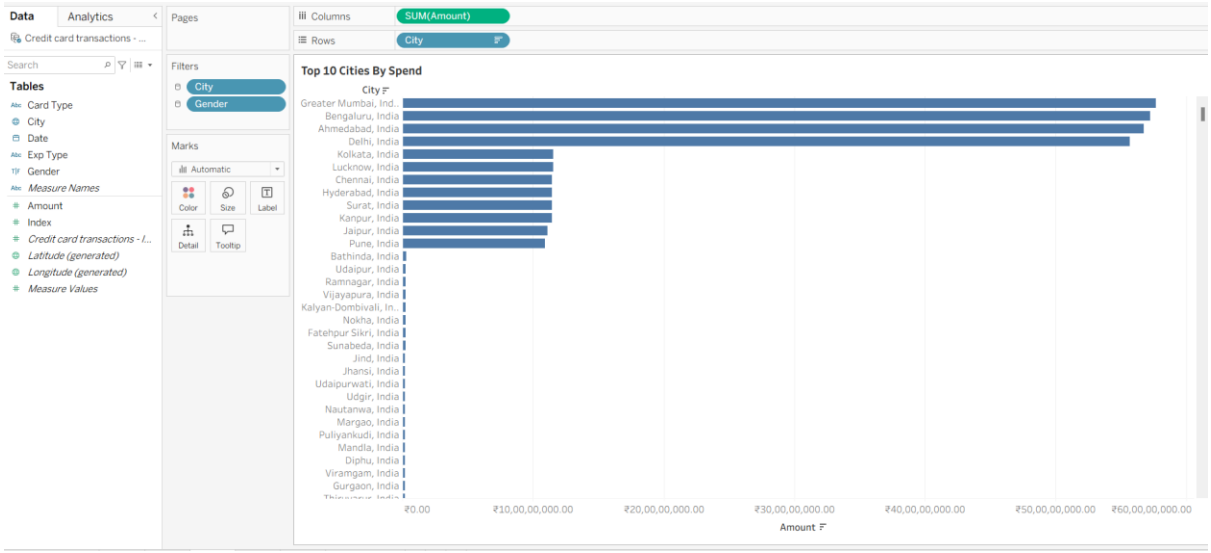
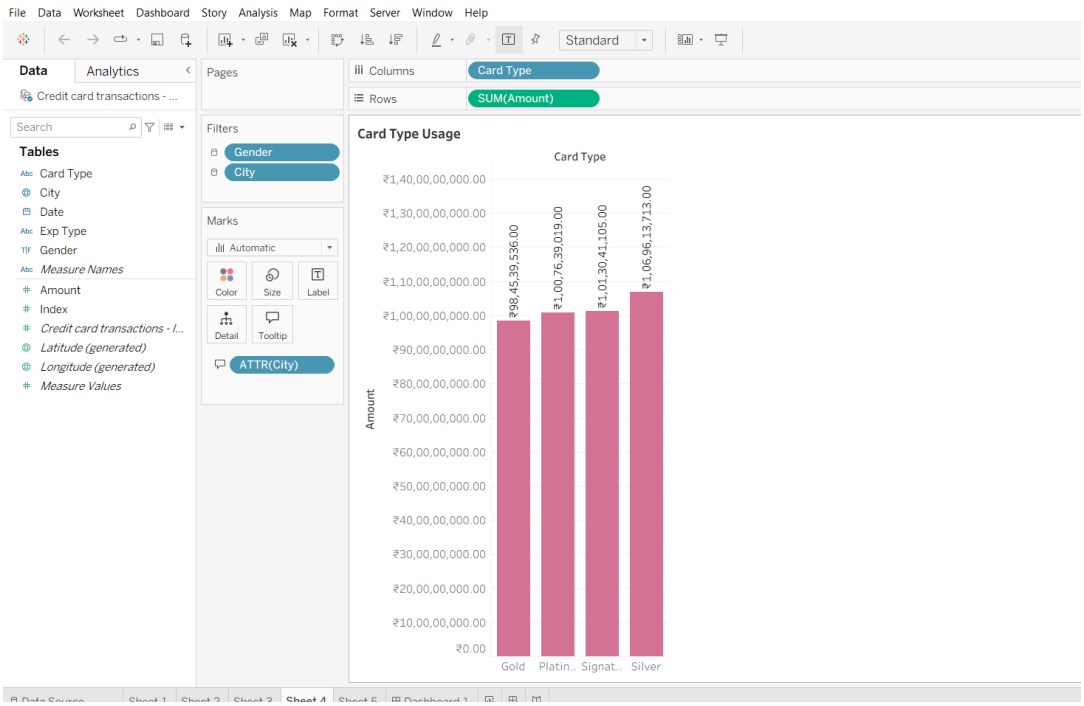


4. Top 10 Spenders

Horizontal bars ₹1-5 lakh.



5. Total Spend KPI

₹36+ lakh gauge.

INTERPRETATION

Total Spending Overview

- Overall spend ₹36-39 lakh range mein hai (KPI gauge se clear)
- High-value transactions focus hai, small spends bhi track hote hain

Expense Type Analysis

- 4 major segments (Auto, Bank, etc.) different colors mein stacked bars
- Har segment ka contribution visible - koi ek dominate nahi kar rha

Gender-wise Spending Patterns

- Male vs Female comparison bar chart se clear difference
- Specific categories mein gender preference dikhti hai

Top Performers Identification

- Top 10 spenders list (₹1-5 lakh range) pink/blue horizontal bars
- Business ke liye high-value customer targeting ke liye perfect

Key Business Insights

1. **Customer Segmentation**: Auto/Bank wale segments highest spenders
2. **Gender Targeting**: Marketing campaigns gender-wise customize kar sakte hain
3. **VIP Focus**: Top 10 customers pe special offers/rewards
4. **Category Performance**: Kaunsa expense type zyada revenue deta hai clear

Technical Highlights (Viva ke liye)

- **Interactive Filters**: Date, segment select karke drill-down
- **Multiple Chart Types**: Stacked bar, horizontal bar, comparison charts
- **Color Coding**: Consistent color scheme across charts

- **KPI Dashboard**: Single glance mein business metrics

KEY INSIGHTS:

The following important insights were obtained from the analysis:

- Most spending in top categories like Auto, Bank via stacked charts.
- Gender differences in expense types visible in bars.
- Top spenders dominate total via pink bars.
- Total spend around ₹36-39 lakh; high-value focus.

CONCULSION

This project successfully analyzed credit card transaction data using Tableau. Revealed insights on spending behavior, categories, and trends.

Highlights data analytics role for financial decisions

Limitations of Study

Some limitations of this study include:

- The Dataset size limited
- Some data missing/inconsistent
- No real-time analysis

Future Scope

This project can be extended in several ways:

- Machine learning for fraud prediction
- Real-time dashboards
- Larger datasets integration

REFERENCES

1. Tableau Official Website: https://www.tableau.com
2. Dataset: https://www.kaggle.com/datasets/thedevastator/analyzing-credit-card-spending-habits-in-india